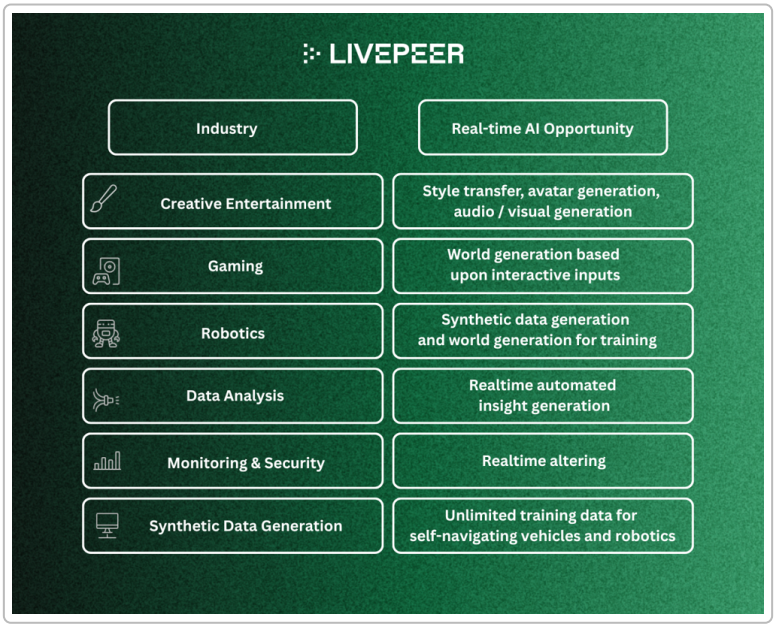
ChatGPT

Livepeer Overview



Livepeer is a decentralized live video and AI streaming network built on Ethereum. It enables developers and creators to add live or on-demand video (and AI-driven effects) to their apps, while scaling to millions of viewers. Broadcasters pay for encoding jobs in ETH (or stablecoins), whereas the Livepeer Token (LPT) is only *staked* by network nodes and delegators to secure and scale the network. Node operators (Orchestrators) must stake LPT to join the network and bid on work [1]. The more LPT staked to a node, the more transcoding and AI work it can perform.

<Info>
LPT holders **do not spend LPT on jobs** – LPT is used purely to coordinate and secure the network. Broadcasters pay fees in ETH, which are then earned by Orchestrators (and shared with their delegators). In contrast, Orchestrators and Delegators *stake* LPT to prove their commitment and earn rewards [1]. This stake-for-access model ensures reliable, high-quality streaming services.
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<Tip>
Daydream (AI Video Demo): For example, *Daydream* is Livepeer's open-source real-time AI video app. It transforms live webcam video with a simple text prompt [2]. Daydream is built on Livepeer's decentralized GPU network (using the ComfyStream plugin), and illustrates the shift to interactive AI. Its developers note that as real-time AI usage increases, Livepeer's network throughput and fee generation will grow proportionally [3].
</Tip>

Here's a quick snapshot of the major Livepeer docs:

<Columns cols={3}>
<Card title="Protocol Overview" icon="cog" href="/protocol-overview">In-depth on-chain/off-chain architecture and data flow.</Card>

<Card title="Actor Overview" icon="users" href="/actor-overview">Roles of Orchestrators, Delegators, Gateway operators, and more.</Card>
<Card title="Governance & Treasury" icon="gavel" href="/livepeer-governance">LIPs, voting, the treasury, and the Livepeer Foundation.</Card>
<Card title="Token Economics" icon="dollar-sign" href="/livepeer-token-economics">LPT staking model, inflation, and reward distribution.</Card>
<Card title="Protocol History" icon="book-open" href="/livepeer-whitepaper">Historic evolution (Streamflow, Cascade, AI pivot, etc.).</Card>
</Columns>

Suggested Media: Daydream livestream demos, ComfyStream workflow videos, and examples like MetaDJ's Daydream performance (video) or Matt Shumer's tweets on Cascade. Also consider diagrams of the protocol flow or token emission (possibly via Mermaid) to illustrate these concepts.

Protocol Overview

Livepeer's protocol consists of off-chain video processing and on-chain coordination. Broadcasters submit video streams, and Orchestrators (video nodes) transcode or apply AI filters to those streams off-chain. Meanwhile, essential coordination (staking, bonding, reward claims, parameter updates) happens on-chain. This split allows high-throughput, low-latency video work without burdening the blockchain.

<Tabs>
<Tab title="Off-Chain Workflow">
Broadcasters (streaming apps) send live or recorded video to the network. Each video segment is accompanied by probabilistic "tickets" (lottery-style micropayments) that encode the requested work and price. Orchestrators process the stream with low-latency GPUs and return the result to the broadcaster. The orchestrator collects these tickets off-chain. Periodically, winning tickets are submitted on-chain to redeem ETH from the broadcaster's deposit [4]. This means live video can flow continuously while payments are batched via a lottery mechanism.

Key Points:
- **Probabilistic Payments:** Each segment of video carries a "ticket." Winning tickets (determined by the orchestrator's price and video length) can be claimed on-chain for ETH [4]. This design keeps the network scalable and gas-efficient.
- **Rewards:** In addition to ETH from tickets, new LPT tokens are minted each round as staking rewards, split between the orchestrator and its delegators.
- **Example Workflow:** A broadcaster locks up ETH in a deposit contract and streams video. Orchestrators attach tickets to each chunk. At the end, the broadcaster's deposit funds the winning tickets, which orchestrators claim as ETH. Delegators earn a share of both the ETH fees (if their node chooses to share) and LPT issuance [5] [4].
</Tab>

<Tab title="On-Chain Coordination">
On-chain, Livepeer manages staking, bonding, and protocol rules. Node operators **stake LPT** to become Orchestrators and set their service price. Each round (epoch), the network elects a new Validator set of top-staked nodes. The Ethereum smart contracts enforce these rules and distribute rewards.

Key Elements:
- **Staking & Delegation:** Any LPT holder can delegate tokens to an orchestrator, increasing that node's weight. This helps decentralized security and aligns incentives [6]. Delegators earn a slice of that node's ETH fees and LPT rewards. (Delegating also gives voting power in on-chain governance.)
- **Bonding and Slashing:** Orchestrators lock LPT as a bond. Misbehavior (e.g. producing bad video or downtime) can incur slashing, penalizing both the operator and its delegators [7]. This keeps service high-quality.
- **Dynamic Emission:** The protocol mints new LPT each round. Livepeer uses a dynamic inflation model: if less than 50% of LPT supply is staked, inflation (and rewards) increases to encourage participation; if above 50%, inflation tapers [8]. This adaptive incentive helps balance security and dilution.

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</Tabs>

<Info>
Livepeer's on-chain contracts require a **minimum of 100 LPT staked** to submit a governance proposal, and votes need at least 33% of staked LPT participating with >50% approval [9] [10]. This ensures stakeholders guide protocol changes responsibly.
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Actor Overview

Livepeer's ecosystem involves both **protocol actors** (on-chain roles) and **community actors** (network participants and builders). Below is a summary of the major actors:

Protocol Actors

- **Orchestrators:** Operators running Livepeer nodes (software) that perform video transcoding and AI workloads. Orchestrators stake LPT to secure the right to serve jobs. They compete on price, latency, and quality. Successful work earns them ETH (from ticket redemption) and LPT rewards.
- **Delegators:** LPT holders who delegate their tokens to one or more orchestrators. Delegators share in their chosen operators' rewards (both ETH fees and minted LPT) [11]. By delegating, they "vote" for reliable nodes and earn passive yield. This strengthens decentralization, as even non-technical users can support the network's security [6] [11]. (Delegated tokens also carry voting power in governance proposals.)
- **Broadcasters:** Applications or users who send video streams to Livepeer for processing. Broadcasters deposit ETH on-chain and consume the network's video/AI services off-chain. They pay for transcoding in ETH via the ticket mechanism. Broadcasters do not use LPT to pay; LPT is only used for staking by the work providers.
- **Viewers:** End-users who watch the live or recorded streams that Livepeer processes. They typically are oblivious to the blockchain; they just see the video output (with any AI transformations applied).

Ecosystem Actors

- **Gateway Operators:** Teams or entities that run "gateway" servers to simplify network access for developers. For example, the **Livepeer Studio Gateway** is a managed API service that routes video jobs to the network [12]. The **Livepeer Cloud Community Gateway** provides a free entry

point. Gateway operators handle things like matchmaking and monitoring, making it easy for apps to integrate without running their own node.
- **AI/Media Builders:** Creatives and developers building on Livepeer's infrastructure. For instance, *Matt Shumer's ComfyStream* project extends ComfyUI (a popular AI workflow tool) to real-time video. ComfyStream (used by Daydream) represents an open-source stack for live AI video effects [13]. Other builders include artists like MetaDJ (who used Daydream for a live performance) and teams building streaming platforms (e.g. Retake.tv for tokenized livestreams). These actors innovate on top of Livepeer's core protocol to create applications and community content.

<Tip>
Delegation is crucial: By **delegating LPT** to high-quality Orchestrators, you help secure the network and **earn rewards**. Delegators receive a share of both the protocol's emissions and actual usage fees [11]. Even as Livepeer's inflation decreases over time, delegators benefit directly from real video/AI workloads run on the network [5] [11].
</Tip>

<Tip>
Gateway nodes simplify access: For example, the **Livepeer Studio Gateway** offers a production-ready API for building video apps without managing your own transcoder nodes [12]. It's a service layered on top of the core protocol, provided by the ecosystem.
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Livepeer Governance

Livepeer uses a hybrid on-chain/off-chain governance model:

- **Livepeer Improvement Proposals (LIPs):** Formal design documents (like Ethereum's EIPs) are drafted to propose protocol changes. The LIPs repo (on GitHub) is where new protocol upgrades are defined. Example LIPs include treasury creation (LIP-89) and voting refinements.
- **Community Voting (SPEs):** The on-chain treasury and protocol upgrades are handled via *Special Purpose Entities* (SPEs). Any community member can draft a pre-proposal (RFP) to build an SPE that implements an idea. Once an SPE team is formed, they submit a proposal to the Livepeer DAO. Proposals must be backed by at least **100 staked LPT** and await a community vote.
- **On-Chain Voting:** Livepeer uses snapshot-style voting on-chain. A proposal passes only if ≥33% of staked LPT participates and >50% of votes favor it [9] [10]. This ensures that major stakeholders have a say. If a proposal passes, funds from the treasury can be allocated to new SPEs or changes can be executed.
- **Treasury:** Livepeer has an on-chain treasury (funded by inflation). Initially 10% of minted LPT per block was set aside for community funding [14]. (When the treasury hit its initial cap, contributions paused – a future LIP is proposing to resume a 10% rate [15].) Funds in the treasury are used to pay for public goods, grant developers (SPEs), and strategic initiatives.
- **Livepeer Foundation:** In 2025, the neutral Livepeer Foundation (LF) was launched to steward growth [16]. The LF is a nonprofit that coordinates long-term vision, core development, and ecosystem support [16] [17]. It facilitates advisory boards and helps align stakeholders, while the tokenholders maintain ultimate authority via on-chain votes.

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Anyone proposing an on-chain change must stake 100 LPT. If the proposal meets quorum (33% of staked LPT) and a simple majority, it passes [9] [10]. Unsuccessful proposals remain eligible for revision. This structured process (inspired by LIP-89, -90, etc.) provides transparency: ideas move from forum

discussion into formal proposals and then on-chain voting [9].
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The Livepeer Foundation was created to accelerate this process. It acts as a credibly-neutral coordinator for strategy and development [16]. For example, in Q2 2025 it launched advisory boards and a public roadmap, helping channel community input into actionable plans [18] [17]. However, all spending and upgrades remain accountable to tokenholders via the DAO process.
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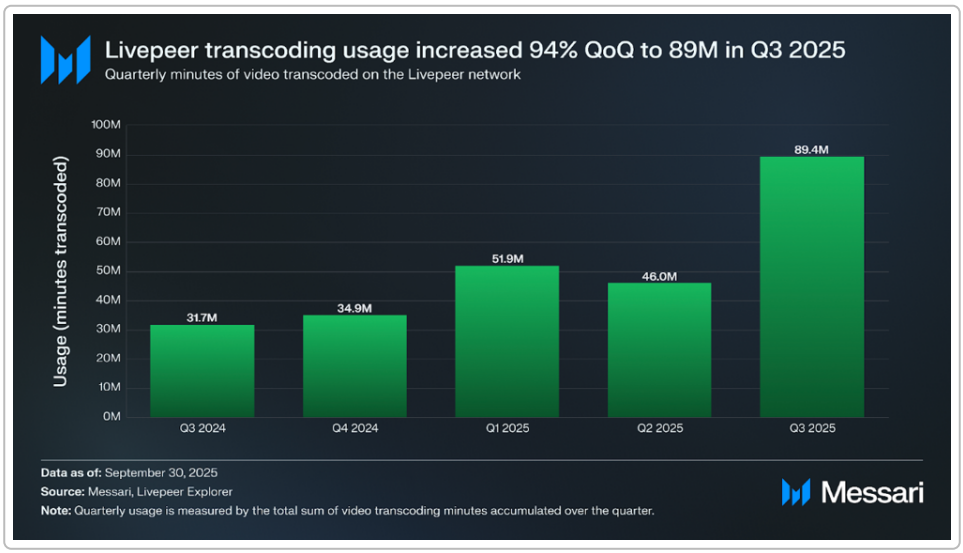
Related Resources: The Livepeer LIPs repo lists all formal proposals. The forum's Governance category has community discussions and working group updates. The Livepeer Foundation's site (livepeer.org/foundation) also shares strategic plans and community news.

Livepeer Token Economics

The Livepeer Token (LPT) is a **stake-for-access** token. Its primary role is to secure the network: node operators must stake LPT to earn the right to transcode video and AI workloads [1]. LPT **itself is not used to pay for jobs**. Instead, broadcasters deposit ETH into a bonded deposit, and the network's earned ETH fees are distributed to Orchestrators and Delegators [19]. LPT aligns economic incentives:

- **Securing the Network:** By staking LPT, Orchestrators prove they have "skin in the game" and can be penalized (slashed) for misbehavior. Delegators amplify network security by delegating to reputable nodes [7].
- **Issuance & Rewards:** Livepeer mints new LPT each block. By default, *90%* of this issuance is allocated to stakers (orchestrators & delegators), and *10%* to the community treasury (until any cap is reached) [20]. These newly minted tokens are awarded to nodes as round rewards. For example, in 2025 annual inflation has been around 25–27% (adjusted dynamically) [8].
- **Usage Fees:** In addition to LPT inflation, Orchestrators earn ETH from broadcaster payments. Over time, Livepeer is shifting toward fee-based rewards: as of late 2025, a majority of network revenue came from actual video/AI job fees rather than new tokens [21] [11]. Delegators benefit by taking a cut of those fees (if their operator opts to share).



A chart of Livepeer usage highlights recent growth: daily video minutes have surged, driven by AI workloads and streaming.

```
flowchart TD
    subgraph "Livepeer Token Flows"
    A[LPT Inflation] -->|~90%| B[Stakers (Orchs & Delegs)]
    A -->|~10%| C[Treasury (funding SPEs)]
    end
```

The above diagram shows that ~90% of each inflation round goes to participants, and ~10% to the on-chain treasury [20]. (Note: the actual percentages can be adjusted via governance LIPs.)

<Info>
Payment in ETH: Broadcasters pay fees in ETH (or a stablecoin) to the network [4]. Orchestrators earn these fees by redeeming winning tickets. Delegators receive a share of those fees and the new LPT emissions in proportion to their stake. Over the last few quarters, Livepeer's fee generation has been growing rapidly, especially from AI video jobs [3] [21].
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Inflation Model: Livepeer's inflation is **adaptive**. If less than 50% of LPT is staked, inflation will increase to attract more stake; if above 50%, inflation decreases [8]. For example, when staking recently crossed 50%, on-chain emissions began tapering. This dynamic system (sometimes called AISP) is designed to balance security incentives and token dilution over time.
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Chart Data: Transcoding usage (minutes) Q3 2024–Q3 2025 (Messari) [22].

Livepeer Whitepaper

The Livepeer protocol has evolved significantly since its inception. The original whitepaper (2017) proposed a decentralized video transcoding network. Over **7+ years**, Livepeer has iterated through many phases [23]: - **2018–2019:** Testnet and mainnet launch. Early incentives were established for Orchestrators to participate.
- **2020:** Stablecoin (DAI) support added for predictable payments. The probabilistic ticket system (micropayments) was rolled out for efficient payments [4].
- **2021:** Ethereum upgrades and dynamic gas pricing introduced. The protocol added a reputation-based winner-selection and replaced some initial gas-heavy operations with off-chain protocols (Streamflow initiative).
- **2022:** Focus on scaling and decentralization. Innovations included upgrades to the bonding curve, governance improvements, and work on separating video and data planes.
- **2023:** The on-chain community treasury was formalized (LIP-89) [14]. The protocol adopted the dynamic inflation mechanism: issuance rises when stake is low and falls when high (a built-in Adaptive Incentive mechanism) [8]. This marked a shift toward balancing token issuance with real network usage.
- **Late 2024:** *Cascade Vision* launched [24]. Livepeer pivoted to become the leading platform for **real-time AI video pipelines**. The Cascade proposal recognized that Livepeer's existing live-streaming stack

(handling 78–92% livestream traffic [24]) is ideal for AI workloads. This spurred development of new tools (e.g. ComfyStream) and community focus on interactive AI video.
- **2025:** The first AI applications hit the network. Livepeer's *ComfyStream* (with ComfyUI) became core to Daydream, an app transforming live video via AI prompts. The Livepeer Foundation was launched to guide growth [16] [17]. With on-chain governance in place, Treasury funds and grants are being deployed to fuel this next phase.

```
gantt
    title Livepeer Protocol History
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    section Milestones
    Whitepaper & Testnet       :2018, 1y
    Mainnet Launch             :2019, 1y
    Stable Payments (DAI)      :2020, 1y
    Dynamic Pricing & ERC20    :2021, 1y
    Onchain Treasury (LIP-89)  :2023, 1y
    Cascade AI Vision          :2024, 1y
    Daydream & ComfyStream     :2025, 1y
```

Livepeer continues to build on this foundation. Its early focus on low-latency GPU video has naturally extended into **real-time AI** as models like Stable Diffusion and LLMs emerge. The network's future iterations (AISP adjustments, new SPEs, AI subnets) are all in service of enabling AI-powered streaming at scale [23] [24]. As Livepeer's community often notes, the token economics and open infrastructure aim to align incentives for all participants – from developers and creators to node operators – in this evolving media landscape.

Key Takeaways: Livepeer started as a decentralized transcoding service; over time it has layered in governance and economic mechanisms (SIP protocols) to ensure sustainability. Today, it leverages those lessons to enable a new class of applications: interactive, AI-enhanced video streaming [23] [24]. The protocol's evolution demonstrates a shift from merely providing video compute to becoming a broad AI/media compute platform, while preserving the open, trustless ethos of the original design.

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https://messari.io/report/state-of-livepeer-q1-2025

[2] [3] Introducing Daydream
https://blog.livepeer.org/introducing-daydream/

[4] A Primer on Livepeer's Probabilistic Micropayments | by Livepeer Team | Livepeer | Medium
https://medium.com/livepeer-blog/a-primer-on-livepeers-probabilistic-micropayments-e16788b29331

[5] [6] [7] [11] [21] Why Delegation Still Matters in a Low-Inflation Environment
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[8] [17] State of Livepeer Q2 2025 | Messari
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[9] [10] Livepeer Governance Process - Help: Getting Started - Livepeer Forum
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[12] AI Gateways - Livepeer Docs
https://docs.livepeer.org/ai/builders/gateways

[13] Building Real-Time AI Video Effects with ComfyStream
https://blog.livepeer.org/building-real-time-ai-video-effects-with-comfystream/

[14] Launching The Governance Working Group (GovWork) - Governance - Livepeer Forum
https://forum.livepeer.org/t/launching-the-governance-working-group-govwork/2613

[15] [20] It's time to ACT! Accumulation & the Treasury Ceiling - Governance - Livepeer Forum
https://forum.livepeer.org/t/its-time-to-act-accumulation-the-treasury-ceiling/3153

[16] [18] Launching the Livepeer Foundation - Governance - Livepeer Forum
https://forum.livepeer.org/t/launching-the-livepeer-foundation/2849

[19] Primer
https://www.livepeer.org/primer-new-design

[22] A Real-time Update to the Livepeer Network Vision
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[23] [24] Livepeer Cascade: Leading Platform for Real-Time AI Video
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